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Anteater Poker Version Alpha

User Manual



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Glossary

Texas Holdem: The standard poker variant anteater poker is based on, where each player holds two private cards and shares five community cards

Community cards: Cards placed faced up on the table that all players get to see when forming their best hand

Check: A poker action that passes the turn without placing a bet, only available when no bet has been made yet

Call: A poker action in which a player matches the current bet to stay in the game

Fold: A poker action in which a player discards their hand and exits the current round, losing all bets they placed during the round

Raise: A poker action in which a player increases the current bet above the previous amount

Pot: The total point accumulated from all players bets in a given round, awarded to the winner

Side pot: A separate pot created when one or more players are all in and cannot match the full bet of other players; remaining players compete for the side pot independently

Anteater Poker: A name for a texas holdem variant of poker that incorporates special new rules featuring anteater themed incorporations.

Ant: A new card unique to anteater poker decks, which is worth one, and can be combined with other cards to create valuable hands

Anteater: A new card unique to anteater poker decks, which is worth thirteen, and can be combined with other cards to create valuable hands

Anteaters Feast: The rarest hand that trumps all other hands, which is a combination of one anteater of any suit, and all 4 ants in the deck.

Bot Player: A computer controlled player that automatically fills any empty seat at the beginning of the game

Client: The program each player runs on their machine to connect to the game server and interact with the poker session

Port: A numbered network endpoint on the server, anteater poker uses the following range: 10,180-10,189

Socket: A software object representing one end of a tcp network connection used to send and receive data between client and server

TCP: Transmission control protocol; a network protocol that guarantees reliable data between the two sockets

GUI: Graphical user interface; the visual window players use to see the game

X11 forwarding: An ssh feature that allows graphical windows from a remote server to display on the users local machine

1. Online Poker

1. Usage scenario

The Anteater Poker game (based on the Texas Hold'em version of Poker) is a game where multiple players can join together and play, in addition to nonhuman players that fill in untaken seats.

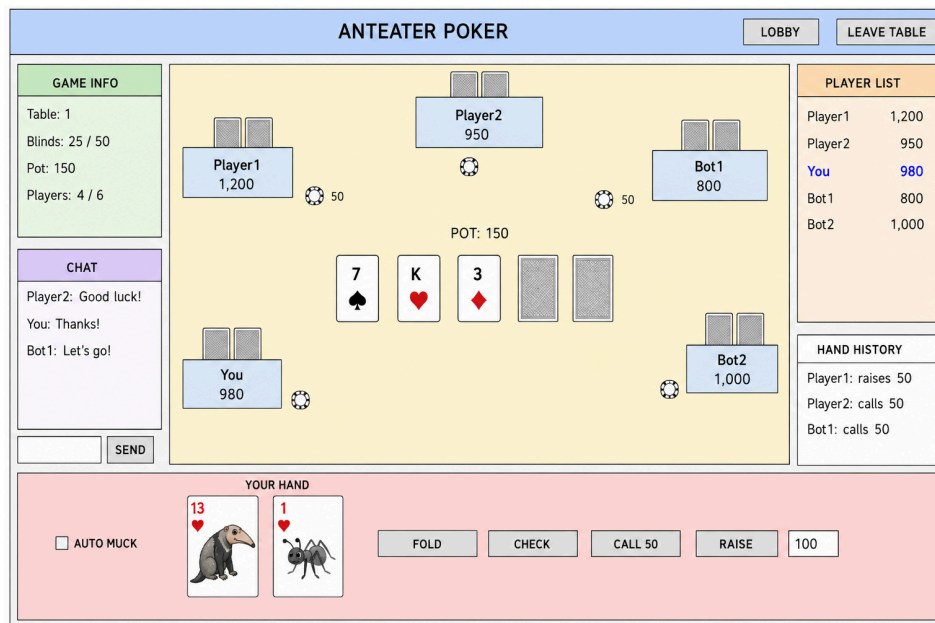
To play the game, the player first launches the client and connects to the Poker game's IP address. After joining the hosted session, the player will select an empty seat, in addition to a username and password of their choosing. When all the players join, the game will start. If some seats aren't filled, bot players will join in the empty seats' place.

Every player is given two private cards, and community cards are put on the table throughout every round of the game. As the game plays out, the players get to choose different actions like check, call, raise, or fold by pressing the button with the indicated action name.

What the GUI will present:

- Player's hand cards
- Community cards
- Player's turn
- Betting amount
- Pot value
- Player scores

When the round ends, the winning hand is determined, and points are awarded to the winner.



Rough Sketch of how gameplay and gui will look

2. Goals

- Connect to the online poker game through the client and be able to play with friends.
- Have a functional GUI.
- Have humans, as well as bot players.
- Play Texas Hold'em poker, but with an Anteater-designed style and mechanic.

3. Features

- Multiplayer gameplay (online)
- Server connection system
- User-Friendly GUI system
- Real time player gameplay
- Texas Hold'em poker gameplay
- Anteater designed and styled gameplay
- Betting System
- Call, raise, check, fold
- Score and pot tracking
- Bot players
- Seat, username, and password selection

2. Installation (15 points)

2.1 System Requirements

Anteater Poker should be run on one of the EECS 22L Linux servers, such as bondi, laguna, crystalcove, zuma, balboa, or doheny. The user will need access to a terminal, gcc, make, and normal TCP/IP socket support. These tools are needed because the game runs with a server and client setup, where the clients connect to the server through sockets.

The project also uses a graphical interface for both the player window and the server control window. Because of this, the system needs to support graphics. If the user is logging in remotely, X11 forwarding may need to be turned on so the game windows can open correctly. The user should also have access to the team account and the project folder named poker.

The release package should include the main program files, executable files, documentation, and folders such as bin and doc. The server should be started first because it controls the main parts of the game, including the deck, turns, points, and game progress. Each player uses the client to enter their name, connect to the server, choose a seat, see their cards, and make decisions like call, raise, or fold.

2.2 Unpacking and Configuration

To install Anteater Poker from the binary release package, first log in to one of the supported EECS servers using the team account. Put the archive file in the directory where the game should be installed. Then unpack the archive with this command:

```
gtar xvzf Poker_V1.0.tar.gz
```

After the archive is unpacked, a folder named `poker` should appear. I would also check that the `bin` and `doc` folders are inside the `poker` folder before running the program. The user manual can be found inside the documentation folder. To open it, use this command:

```
evince poker/doc/Poker_UserManual.pdf
```

The server should be started before any player tries to join the game. Start the server with this command:

```
poker/bin/server &
```

After the server is running, start the poker client with this command:

```
poker/bin/poker
```

Once the client is open, the player can enter their name, connect to the server, and pick an available seat. The server keeps the game running in the background, while the client is mainly used by the player to see their cards and make choices during their turn.

If the user is working with the source code package instead of the binary package, unpack the source archive first:

```
gtar xvzf Poker_V1.0_src.tar.gz
```

Then go into the project folder and build the program:

```
cd poker  
make
```

To run the required tests, use:

```
make test
```

To clean the compiled files after building or testing, use:

```
make clean
```

2.3 Uninstalling

To uninstall Anteater Poker, close the client and server programs first. If the server is still running in the background, stop it from the terminal before deleting the project files.

Go to the directory that contains the `poker` folder. Then remove the folder with this command:

```
rm -rf poker
```

This deletes the installed game files, the executable files, the documentation, and the related project folders from that location. If the archive files are still saved and are no longer needed, they can also be removed:

```
rm Poker_V1.0.tar.gz
```

```
rm Poker_V1.0_src.tar.gz
```

Once the poker folder and the saved archive files are removed, Anteater Poker is no longer installed in that directory.

3. Program Functions and Features (20 points)

3.1. Detailed description of client and server communication

- The poker client will communicate with the poker server via transfer control protocol (TCP). This guarantees consistency in that every bit sent between the client and the server will be delivered without duplicates or errors. To initialize software processes (the poker program) between the client and the server we will connect to one of the UCI hosts that contains our poker program (on allotted port range 10,180-10,189), then utilizing socket communication we will effectively be able to run the server's program through the client.
- The client must first create a socket which will open a one-to-one network connection point for TCP communication with the server. The client will initiate connection with the server while the server constantly waits for someone to request to connect. Once the server accepts the connection the online poker session will begin.
- Throughout this session the client will send different actions such as fold, call, raise, amount, and end game/join game. The server will then respond with states such as cards, pot, whose turn it is, and what game you are in.
- The server handles up to seven clients total, assigning them to the same port in the allotted port range. Each client will be handled separately

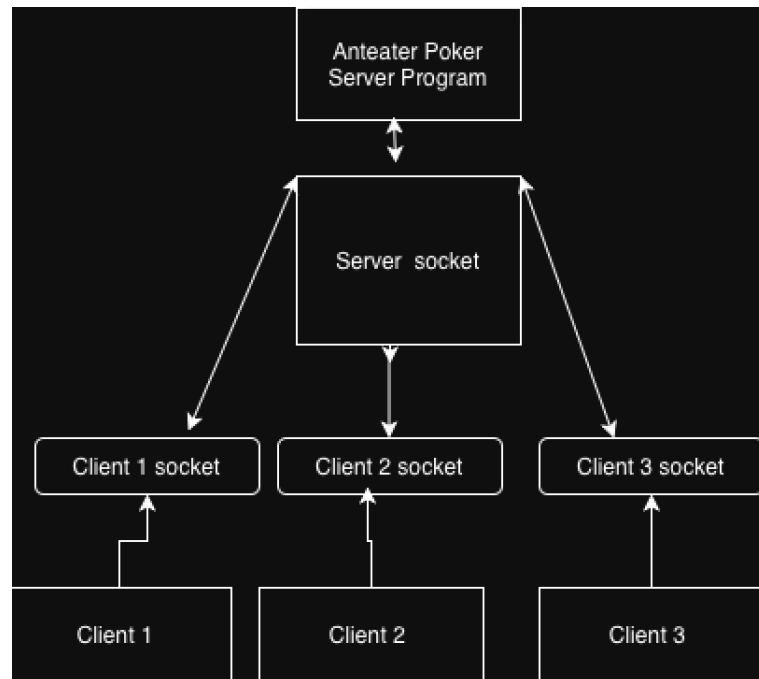


Fig 1. Diagram of network communication between poker clients and poker server

3.2. Detailed description of dealer choice and card distribution

The rules for Anteater Poker are mostly similar to the rules for Texas Hold 'em. The differences in some rules are due to the extra cards. The cards available in Anteater Poker are the regular cards in Texas Hold 'em along with the 'Anteater' and 'Ant' card.

The rules are as follows:

1. Two cards, known as hole cards, are dealt face down to each player, and then five community cards are dealt face up in three stages.
2. The stages consist of a series of three cards ("the flop"), later an additional single card ("the turn" or "fourth street"), and a final card ("the river" or "fifth street")
3. Each player seeks the best five-card poker hand from any combination of the seven cards: the five community cards and their two hole cards.
4. Players have betting options to check, call, raise, or fold.
5. Rounds of betting take place before the flop is dealt and after each subsequent deal.
6. The player who has the best hand and has not folded by the end of all betting rounds wins all the money bet for the hand, known as the pot.
7. In certain situations, a "split pot" or "tie" can occur when two players have hands of equivalent value.

The following hands in order of increasing value are available in the game:

1. High Card - Simple value of the card. Lowest: 2 – Highest: Ace
2. Pair - Two cards with the same value
3. Two Pairs - Two times two cards with the same value
4. Three of a Kind - Three cards with the same value
5. Straight - Sequence of 5 cards in increasing value (Ace can precede 2 or follow up King, but not both), not of the same suit
6. Flush - 5 cards of the same suit, not in sequential order
7. Full House - Combination of three of a kind and a pair
8. Four of a Kind - Four cards of the same value
9. Straight Flush - Straight of the same suit
10. Royal Flush - Highest straight of the same suit
11. Anteater's Feast - Anteater with all four ants

3.3. Detailed description of poker game integration

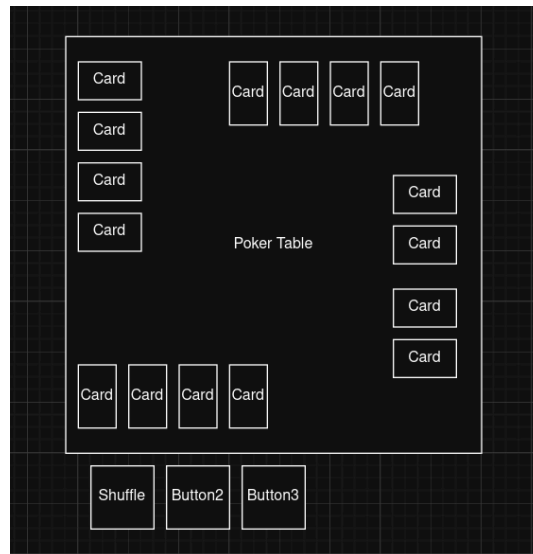
The basic poker features that are offered are:

- The player can connect to the server, reserve a seat at the poker table and choose a username
- The player can interact with the poker application using a GUI. The GUI shows the player their hand, other player's name, and allows the player to interact with their cards.
- Cards in a hand can be randomly shuffled and then dealt to the other players.
- Empty seats are filled with bot players. These bot players can also play and follow the rules of Anteater poker.

The advanced poker features that are offered in the application are:

- A timer is shown that displays how long the current round has been going on for.
- A scoreboard is displayed at the end of the game, showing the player rankings and what poker hands they received.

The basic GUI will look like this:



- The cards of the players will be displayed face down so that fair gameplay occurs.
- The players can see their own cards
- The player can access buttons such as shuffle, sort by rank, and so on

4. Visual Presentation

- For the visual presentation we will use traditional french suited cards for the base 13 cards and for the remaining custom ant and anteater cards we will use our own custom designs as shown below.



Fig 2. Sample depictions of what the additional cards will look like

Back matter

- Copyright

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Anteater Poker was developed as part of EECS 22L: Software Engineering Project in C Language at the University of California, Irvine.

This software and all related documentation, including this User Manual, were created by the team Why So Serious Coders for educational purposes only.

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Third-Party References: Anteater Poker is based on the general rules of Texas Hold'em Poker. Anteater Poker is a custom made variation of the game, created specifically for this project. Redistribution or modification outside of EECS 22L requires permission from the Why So Serious Coders team and course staff.

- Error messages

“Unable to connect to server” - Client cannot establish a connection to the server

“Connection timed out” - Server did not respond within the expected time

“Seat already taken” - Seat is currently being occupied by another player

“Invalid username or password” - Login information entered is not accepted

“Game is already in progress” - A round is already underway and no new players can join until the next round

“Not your turn” - Player attempted an action when another player is active

“Insufficient points to raise” - Player does not have enough points to make the requested raise

“Invalid bet amount” - Betting value is not allowed

“Server disconnected” - Server was closed or connection was lost

“Bot initialization failed” - Computer-controlled player cannot be created

“Deck synchronization error” - Client and server card states do not match

“Graphics display unavailable” - GUI cannot open because graphical support is missing

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